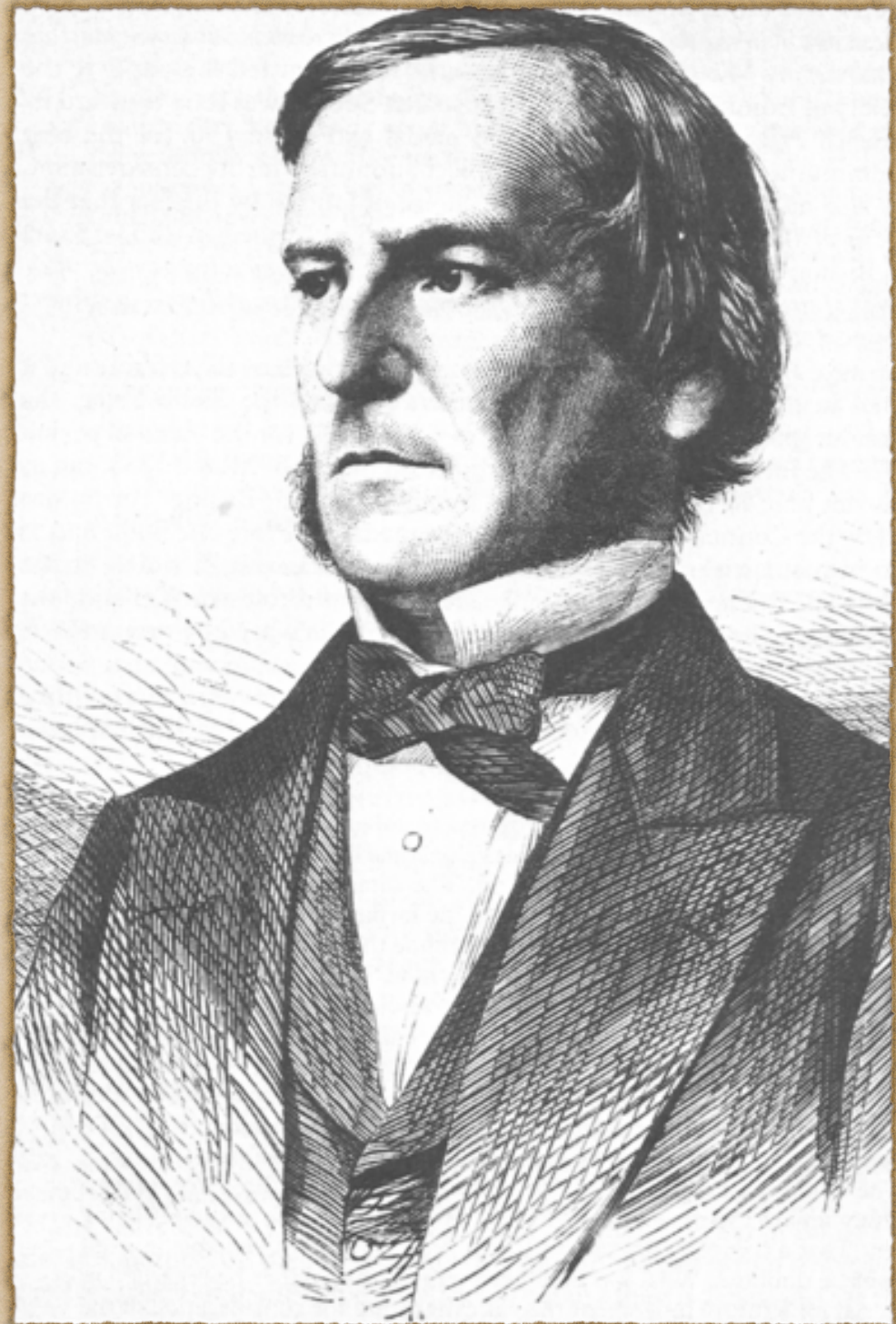


Lecture 6

Evan Drake

Boolean Logic



https://en.wikipedia.org/wiki/George_Boole

Systems of Logic

- ◆ Truth seeking
- ◆ Formal logic
- ◆ System of creating proofs

Other logical systems

- ◆ Lambda Calculus
- ◆ Predicate logic
- ◆ Propositional logic

"Much of my work has come from being lazy."

—John Backus

The Basis

Bool	True (T)	False (F)
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- ◆ The Boolean

Operations

- ◆ Not
- ◆ And
- ◆ Or
- ◆ XOr
- ◆ NAnd

Unary Operations

◆ Not

A	$\sim A$
0	1
1	0

Binary Operations

◆ And

A	B	$A \wedge B$
0	0	0
1	0	0
0	1	0
1	1	1

Binary Operations

◆ Or

A	B	$A \vee B$
0	0	0
1	0	1
0	1	1
1	1	1

Binary Operations

◆ XOr

A	B	$A \oplus B$
0	0	0
1	0	1
0	1	1
1	1	0

Binary Operations

◆ NAnd

A	B	$\sim(A \wedge B)$
0	0	1
1	0	1
0	1	1
1	1	0

“It does not matter how slowly you go as long as
you do not stop.”

-Confucious

Next week

- ◆ Control Flow
- ◆ Python Practice

Lab

- ◆ Lets work on the Paper